

SESSION BREAK · PAPERS AND DEMOS · SPEAKER SLIDES

3GPP SA4 Papers: 5G/6G Media & AI

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5G-MAG, Fraunhofer FOKUS, BBC, Orange)

On Sunday



IBC2026

FROM 5G MEDIA STREAMING TO 6G MEDIA DELIVERY: LESSONS LEARNED AND FUTURE DIRECTIONS IN 3GPP

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ABSTRACT

5G Media Streaming (5GMS) defines a service-based architecture that decouples media applications from the specifics of transport, access, and network deployment. This architecture supports downlink and uplink streaming, edge distribution, dynamic policies, and network assistance features while integrating with standard content delivery ecosystems like HTTP adaptive streaming. Building on initial deployments and trials of Advanced Media Delivery (AMD) features in 5G (Release 19), the present paper details the Advanced Media Delivery Phase 2 (AMD Phase 2) enhancements under development in 3GPP. These features refine the architectures of the 5GMS System and 5G Multicast–Broadcast User Services to improve observability, robustness, and network–application collaboration, with minimal service-specific complexity. This includes expanded in-band client metrics reporting (e.g., Common Media Client Data v2), pragmatic network assistance via standardised signals (e.g., SCONE bit rate guidance and low-latency ECN marking), and interoperable unicast–broadcast service combination. The role of energy efficiency in media delivery is examined, leveraging emerging 5G mechanisms for energy consumption exposure and adaptation. Additionally, open-source 5G-MAG Reference Tools are highlighted as an enabler for validating and adopting these advanced features. Finally, an outlook on 6G Media Delivery is provided, emphasising an evolutionary path that reinforces continuity, architectural harmonisation for streaming and real-time services, enhanced network observability, and integrated compute capabilities, while preserving simplicity, deployability and interoperability.

15:45 –

17:00

RF Networks and Distribution Developments

📍 Conference Room 2

This session will discuss
developments in RF networks
and distribution technologies.
The session will ...

Moderator



Phil Kesby, Domain Architect
– BBC Architecture - Arqiva

Speakers



Joe Fabiano, CTO -
Edgebeam Wireless, LLC



Phil Brown, Senior Principal
Engineer - Spectrum - Arqiva



Rafael Mekuria, Principal
Engineer - Huawei



Thomas Stockhammer,
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Standards - Qualcomm



IBC2026

AI FOR 6G AND 6G FOR AI – INITIAL FINDINGS FROM 3GPP

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ABSTRACT

Artificial Intelligence (AI) and 6G are emerging as deeply intertwined pillars of future communication systems. On the one hand, AI is expected to become a major class of 6G services; on the other hand, 6G networks must themselves evolve to efficiently support AI workloads. This paper explores this duality—AI for 6G and 6G for AI—from the perspective of ongoing work in 3GPP SA4, which focuses on media systems, codecs, and characteristics. Initial 3GPP studies highlight that AI-generated and AI-consumed data (real-time conversational AI, multimodal perception, neural network–based codecs, federated learning) differ significantly from traditional media traffic in terms of latency sensitivity, burstiness, uplink/downlink asymmetry, and tight coupling between generation and communication. Understanding and characterizing such traffic has therefore become a key building block for 6G system design. AI-assisted media streaming, adaptive coding, traffic prediction, and service-aware optimization offer new opportunities to improve network efficiency, quality of experience, and energy consumption. This paper builds on prior work in 6G media [1], extending it with new insights on AI-native networks, AI-centric traffic modeling, user-experience-centric KPIs, and ecosystem development via open tools. We present an evidence-based methodology for characterizing media traffic and share initial results from a cross-industry 6G media testbed, highlighting early findings that can guide the design of 6G services and networks that treat AI-driven applications as first-class citizens.

Terms— 6G, media streaming, AI-native networks, traffic characterization, immersive quality of experience (QoE), non-terrestrial networks, generative AI, content delivery, open ecosystem.